

The University of Lahore
Department of Computer Science & IT
Mid Term Examination Solution



Spring-2024

Course Title:	Object Oriented Programming	Course Code:	CS02203	Credit Hours:	4
Course Instructors:	Dr.Mulahammad Atif, Mr. Ali Raza, Ms. Iqra Adnan, Ms Haseena Kainat, Mr. Sardar Waqar Khan	Program Name:	BSCS		
Time Allowed:	60 Minutes	Maximum	25		
Date:	18-03-2024	Course Mentor:	Dr. Muhammad Atif Chattha		
Student's		Signature:			
Reg. No:		Section:			

Note:

- 1) It is an open book exam.
- 2) Understanding questions is part of examination, so no queries will be answered.

Question # 1: (10 Marks)

Design a class called "Laptop" to model laptop devices. Each instance of the "Laptop" class should possess attributes representing its brand, model, and owner's name. Implement a copy constructor for the "Laptop" class, which initializes a new laptop object with the same brand and model as the original object, but with an empty owner's name. Additionally, provide a setter method named "setOwnerName" to allow the assignment of an owner's name to a laptop object after its creation.

In the main function, demonstrate the utilization of the "Laptop" class by performing the following steps:

1. Create an original instance of the "Laptop" class.
2. Copy the original laptop object to a new laptop object.
3. Utilize the "setOwnerName" method to assign an owner's name to the new laptop object.
4. Display the information of each laptop, including brand, model, and owner's name.
5. Calculate and display the average price of the laptops.

You have the freedom to determine whether additional constructors or methods are necessary to achieve the desired functionality..

Question # 2: (15 Marks)

Design a class "Rectangle" with attributes length and width. Create a derived class "PlayArea" from "Rectangle" and add an attribute "surfaceType". Additionally, create another derived class "PlayGround" from "PlayArea" with an attribute for the number of players. In the main function, instantiate three objects: football ground, lawn tennis ground, and basketball ground. Display the data of the playground with the maximum area. Also, calculate and display the average number of players across all grounds. You have the discretion to decide whether additional constructors or methods are required for functionality where no data member is public.

Good Luck